



Mehran Rajabi

Software & AI Engineer | M.Sc. Intelligent Adaptive Systems (IAS)

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Summary

Full-Stack Software and Data Engineer with **3+ years of experience** in **Python, C#/.NET**, and **Java**. Master's student in Intelligent Adaptive Systems at **UHH**, focusing on **Machine Learning** and **Computer Vision**. Combines scalable **backend architectures** with **data pipelines** and productive **LLM/RAG integration** – complemented by **DevOps** practices (**Docker, CI/CD**, Cloud).

Work Experience

- Working Student – Software Engineering** – *comserro GmbH, Norderstedt / Hamburg* 10.2026 – Present
- Development of **cloud security portal architectures** (M365) using an **AI-first development approach** (LLMs); integration of **Microsoft Graph API** and **REST APIs** for modular backend and frontend features.
- Full-Stack Developer** – *Shahrekord CO2, Iran* 05.2022 – 10.2025
- Full-stack implementation of an industrial **ASP.NET Core** IoT platform: frontend (**React.js**), high-performance backend (**REST**), and relational database (**Microsoft SQL Server**), focusing on high maintainability and Clean Code.
 - Architecture of robust **data pipelines (ETL)** processing over **80 sensor data records per second** from **>100 distributed hardware endpoints**, designed for cloud migration.
- Software Developer** – *Radshid, Isfahan, Iran* 12.2020 – 03.2021
- Co-led the development of a **real-time tracking application** (React.js / **C#**) processing telemetry data from **over 1,000 active sensors** and vehicles.
 - Backend refactoring** (Onion Architecture, OOP) and development of an automated data service, optimizing system resources and **saving 10% of storage space**.

Software, AI & Data Engineering Projects

- TRUST – AI-driven Supply Chain Risk Analysis** 🔗 – *Hamburger Hochbahn AG* 2026
- End-to-end realization of a compliance prototype (**Python/Flask, Docker**) with an integrated **RAG pipeline**, extracting unstructured data from **50+ web sources** and making it **queryable in under 5 minutes**.
 - Integration of **LLMs** (GPT-5.2, LangChain) for product teardown (**>90% precision, -40% input errors**).
- Behavioral Cloning Framework (Deep RL)** 🔗 2026
- PyTorch** framework for training visual (**ResNet**) & state-based offline policies, including **HDF5** data processing.
- Computer Vision & Predictive Modeling** 2023 – 2024
- ML projects** utilizing **TensorFlow, PyTorch, Pandas, NumPy**: **LeNet-CNN** for smile detection (**+11% gen.**) 🔗, **SVM pipeline** 🔗, **K-Means** color segmentation 🔗, and **KNN model** for revenue forecasting 🔗.
- Automation & IoT Tools** 2022 – 2024
- C#/.NET CLI tool** for **LLM-assisted flashcard generation** (Gemini, JSON) – **>95% time savings** 🔗; **IoT solution** (ESP32) for climate control 🔗.

Education

- M.Sc. Intelligent Adaptive Systems / Artificial Intelligence** – *University of Hamburg, Germany* 10.2025 – Present
- Specialization in **Artificial Intelligence**, focusing on **Machine Learning, Neural Networks**, and **Computer Vision**.
- B.Sc. Information Technology** – *ACECR Institute of Higher Education, Isfahan, Iran* 01.2015 – 02.2020
- Fundamentals of software engineering and modern IT systems using **Java, C#, ASP.NET, C++**, and **SQL Server**.

Skills

Programming Languages	Python, C#, Java, TypeScript/JavaScript, C++, PHP, SQL, Bash
Software Eng. & Web	Full-Stack, OOP, REST API, ASP.NET Core, Flask, React.js, Entity Framework, Testing & QA
DevOps & Cloud	CI/CD Pipelines (GitHub Actions, GitLab CI, Jenkins), Docker, Kubernetes, Terraform, Azure/AWS/GCP, Linux
AI & Data Science	Machine Learning, Deep Learning, Computer Vision, RAG/LLM, Data Analytics, Data Quality
ML & Data Libraries	TensorFlow, PyTorch, LangChain, Scikit-learn, OpenCV, Pandas, NumPy
Databases	Microsoft SQL Server, PostgreSQL, SQLite, Vector Databases
Teamwork & Methods	Scrum & Agile, Jira, Notion, MS Teams, Git, Code Reviews, team-oriented communication
Soft Skills	Analytical Mindset, Teamwork, Structured Problem-Solving, Fast Adaptation to New Tech Stacks

Certificates

Course on Artificial Intelligence – Mehregan Inst. 🔗 Java 🔗 (ACECR) | PHP 🔗 (Sololearn)

Languages

Persian (Farsi) – Native English – B2 (IELTS 6.5) 🔗 German – B1 (actively learning)